Rami Reddy Sarimella

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.40
Narayana Junior College	Class XII (Board of Intermediate Education, Andhra Pradesh)	2017	93.60%
Blue Bells Public School	Class X (Board of Secondary Education, Andhra Pradesh)	2015	9.50%

SCHOLASTIC ACHIEVEMENTS

Secured gold medal in B.Tech as Branch Topper

POSITIONS OF RESPONSIBILITY

- Vice captain and Design Engineer in SAE Baja 2020
- Telugu Club - Head of Teaching Department

SKILLS

- Catia
- Mujoco
- ROS2
- Fusion 360
- Ansys Mechanical

RELEVANT COURSES

- Fundamentals of Robotics
- Autonomous Robotic Systems
- Control systems Design
- Vibrations
- Advanced Mechanics of Solids

PROJECTS

- Experimental Setup for Obstacle Crossing in Prismatic Knee Biped using Reinforcement Learning
- Fabricated a prismatic knee biped robot and supported planar walking using RL and Genetic Algorithms.
- Utilized SolidWorks and 3D printing to validate obstacle crossing heights and widths.
- Autonomous Mobile Robot with ROS2
- Developed a 4-wheel differential drive robot using ROS 2, YDLIDAR, and IMU for autonomous navigation.
- Integrated multi-sensor perception (RGB camera, LiDAR) for a research paper on autonomous semantic based navigation.
- Vibrations and Dynamics
- Measured viscous damping in piston-cylinders using Laser DAQ and Python-based OMI algorithms.
- Validated results against analytical models, demonstrating a high patent-potential methodology.
- Fabrication of a Wheeled Humanoid Robot for Human-Robot Interaction
- Fabricated a cost-effective wheeled humanoid robot optimized for HRI research applications.
- Developing interactive capabilities by integrating local Large Language Models (LLMs) within the ROS2 middleware ecosystem.

EXTRACURRICULAR

- Power Lifting
- Running
- Painting